

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.1 **Category:** Petrochemical Engineering

Judge ID#: 146

Department: CHE

Project Title: Production of Low Sulfur Gasoline using HDS

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	4
5. Project innovation	1
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	2
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
38	

Project strong points:

Good Intro Excellent table for the chemicals

Areas for improvement:

Process Flowscheme missing a CFE, then air cooler and then Maybe a water cooler before gas/liquid separator. Amine treating is best done around 100F. Feed back that I have from Contractors is that gasoline demand will fall by 2030 Better modeling of the reaction would be a good idea. The heat rise in this type of process (desulfurization AND Olefin saturation) is usually quite high.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.1 **Category:** Petrochemical Engineering

Project Title: Production of Low Sulfur Gasoline using HDS

Judge ID#: 197

Department: CHE

Technical Quality	
1. Identification of project objective(s)	2
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	2
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	2
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
36	

Project strong points:

Stepped through Process creation well. All steps were detailed

Areas for improvement:

Become more familiar with original Hydroprocessing technology. Background knowledge and process considerations were weak. Also, please be careful in predicting unrealistic profit returns. It is always best to be conservative rather than make grand promises.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 146

Project ID: CHE.10 **Category:** Petrochemical Engineering

Department: CHE

Project Title: Powering India and Helping Africa's Economy through MTO Process

Technical Quality

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	5

Presentation Quality

1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4

Total Overall Score: 48

Project strong points:

Very well done. Good intro of the team

Areas for improvement:

Probably should know more about the Rx-Regen residence time.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 197

Project ID: CHE.10 Category: Petrochemical Engineering

Department: CHE

Project Title: Powering India and Helping Africa's Economy through MTO Process

Technical Quality

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	5
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4

Presentation Quality

1. General professionalism and oral communication	4
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	3

Total Overall Score: 53

Project strong points:

Group was well versed in technologies and motivation for project objective. It was very evident that they spent time in thinking through and gaining understanding of the process. This will be useful for future careers

Areas for improvement:

The presentation quality could have used improvement. Not many visuals. Motivation was a strong point but not made very clear and appealing to the audience.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 146

Project ID: CHE.11 **Category:** Petrochemical Engineering

Department: CHE

Project Title: Para-xylene Production Through Methylation of Toluene

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	2
4. Overall fulfillment of project objectives	5
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
46	

Project strong points:

Excellent presentation, Flow scheme easy to follow

Areas for improvement:

Cost is much too low. Mentors need to provide some rules of thumb to allow students to get a clue if their size/cost is in the correct order of magnitude.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 197

Project ID: CHE.11 Category: Petrochemical Engineering

Department: CHE

Project Title: Para-xylene Production Through Methylation of Toluene

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	2
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	2
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	5
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
42	

Project strong points:

Great visuals. Good indepth understanding of the petrochemical industry, importance of process, and value of chemicals.

Areas for improvement:

Project is missing realist comparisons such as typical equipment specifications.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.12 **Category:** Environmental Engineering

Project Title: Polylactic Acid for our Bioplastic Assets

Judge ID#: 125

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
57	

Project strong points:

1) Using already cultivated crop (sugarcane bagasse) thus no new land need to be cleared. 2) Reduction of greenhouse gases. (Very important project) 3) Reduction in waste. (Very important project) 4) Use for packaging materials for water and beverage bottles 5) Process flow diagram and Block Diagram was very well presented. 6) Operational design safety features were discussed very well.

Areas for improvement:

1) Safety analysis indicated that ignition and flammable issue was covered but all equipment need to be supported seismically for structural integrity and position retention. 2) No discussion was presented for thermal analysis and earthquake analysis. 3) In the Capital Cost Summary table, a line should be shown for maintenance cost and training of personnel so that plant efficiency is maintained. 4) Since the plant will be in Brazil, it will be difficult to control social distancing and other safety regulations in the post pandemic situation in USA as well as in Brazil.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.12 **Category:** Environmental Engineering

Project Title: Polylactic Acid for our Bioplastic Assets

Judge ID#: 135

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
45	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.12 **Category:** Environmental Engineering

Project Title: Polylactic Acid for our Bioplastic Assets

Judge ID#: 142

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	5
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
57	

Project strong points:

I really appreciated the use of cultivated crop waste. Very strong overview and project highlights (purity, output, revenues) and comparison to the industry. Great to hear that the selection process included a consideration of Midwest American corn.

Areas for improvement:

With the market approaching \$4.58 million revenue / K-Ton (\$5.32 billion revenue vs 1162 K-Ton); this approach appears to come to \$1.7 million revenue / K-Ton (\$244 million vs 143 K-Ton). Exploring the reason for this gap (or clarifying why this isn't comparable) would help determining viability.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.12 **Category:** Environmental Engineering

Project Title: Polylactic Acid for our Bioplastic Assets

Judge ID#: 154

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	3
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
41	

Project strong points:

Good group participation.

Areas for improvement:

ROI seems to be very low.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.13 **Category:** Petrochemical Engineering

Project Title: The United States, an Energy Exporter

Judge ID#: 146

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	3
2. Knowledge of background material	2
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	2
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
41	

Project strong points:

Good intro of team in the online presentation Great improvements since the 3rd presentation.. This begins to look like a real design

Areas for improvement:

Be clear on the design basis and objectives... The mercury removal generally is so effective that other treatments are not required.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.13 **Category:** Petrochemical Engineering

Project Title: The United States, an Energy Exporter

Judge ID#: 197

Department: CHE

Technical Quality	
1. Identification of project objective(s)	2
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	3
2. Knowledge of background material	4
3. Organization and quality of presentation materials	2
4. Project description (see project abstract in program book)	1
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	1
Total Overall Score:	
31	

Project strong points:

Succeeded in addressing all the detailed steps of the project.

Areas for improvement:

Presentation was very plain with minimal visual aids. Objective and incentive of project not made clear. Overall communication of project could use improvement.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.14 **Category:** Bioprocess Engineering

Project Title: Human Waste to Biofuel

Judge ID#: 137

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	0
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	2
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
37	

Project strong points:

I think you were quite knowledgeable on the downstream portions of your process. I think your P&ID and PFD are accurate, as well as the economic analysis insofar as some of the design was out of scope.

Areas for improvement:

My main concern with this process is that most of the carbon is in the sludge, but you chose not to evaluate how to get the sludge to the reactor. Your design and economics assumes this happens, but I think it is one of the main challenges of the project.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.14 **Category:** Bioprocess Engineering

Project Title: Human Waste to Biofuel

Judge ID#: 201

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
47	

Project strong points:

Pretty much the only project to suggest a pilot study. A lot of variables are assumed, so this is a prudent measure.
Very concrete impact of project

Areas for improvement:

The title is not completely reflective of the project.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 125

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
50	

Project strong points:

1) It is a revolutionary project - transporting mobile treatment system to the Fracking Locations 2) Fracking has an average of 30% of flowback water returning to the surface. 3) Process steps taken in the project to mitigate the drawbacks encountered in handling flowback water by: 1) Deepwell Injections, which can contaminate ground water aquifers and 2) Open air pits, which can evaporate harmful chemicals into atmosphere. 4) Treating 179 tons/day waste water is very ambitious. 5) Total Revenue of \$12.6 million is also very good. (Comment: this may be impacted if personnel training and equipment efficiency and reliability is not maintained on a continuous basis and on a regular schedule). Is this revenue after the cost of training and maintenance?

Areas for improvement:

1) Consider adding some discussion about the impact on environment using activated carbon. 2) Need to add some steps to ensure that equipment is properly supported in truck to withstand impact due to potential natural hazards like tornado, hurricane and earthquake. 3) The system should also incorporate changes which will come into effect later after current lockdown (COVID-19) considering regulations related to social distancing etc. This will also have impact on estimated days shown (330 days) and 55 sites/year. 4) It is indicated that the equipment will be operated by a Generator. Need to install in the mobile equipment a stand-by generator also, in case of malfunction of the generator.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 135

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
42	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 142

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	5
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	4
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
53	

Project strong points:

Practical solution to a present problem. Well thought out approach to mobilizing treatment efforts.

Areas for improvement:

Missed the chance to ask follow up questions (not at their fault); but I would be curious to ask if these locations have lagoons typically already, or if not, if the mobile treatment would have to align with fracking times from a scheduling perspective. There is a lot of optimism in uptime even with travel days/setup/breakdown times. And the organics waste stream appeared to have a value, what are their markets and what cost is there in transport/delivery? While clean water is always needed, if fracking were legislatively limited or curbed, what would be the viability/market for mobile cleanup year round?

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 Category: Environmental Engineering
Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 154
Department: CHE

Technical Quality	
1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	3
2. Knowledge of background material	3
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
39	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 180

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
52	

Project strong points:

The abstract is very clear and easy to read. The presentation had a good flow. The problem statement is very interesting and the outcomes of this project can have a huge impact on the environment. I am passionate about conserving water and this sounds like a good way to conserve water and lead the world to a hydrated environment.

Areas for improvement:

I am still not sure how and from where you all got the numbers and the information. Did you interview anyone from the on-site (Marcellus Shale) or any related government body to discuss the issue and the solutions? The pie graphs in the ppt could have been drawn using more distinctive colors for a clearer read. I would expect to know about the down time of the treatment setup (though its on trucks). What exact machineries would be used and individual cost and lifespan of the machineries. Overall good work!

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 234

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
50	

Project strong points:

Practical project that needs to be considered. Cost of crude extraction will go up and eventually we will be looking at alternatives again. I think this project has potential but needs more study.

Areas for improvement:

Although, not quite in the scope of the project, but the waste streams need to be looked at (i.e. waste oil and organics)

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.15 **Category:** Environmental Engineering

Project Title: Hydraulic Fracturing's Dirty Secret

Judge ID#: 240

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	2
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
37	

Project strong points:

The subject of the project is interesting and important

Areas for improvement:

I would've liked to see more comparison between the current system and the proposed system. The group also did not really think about where any of the waste materials would go throughout the process and how that may impact the costs or logistics.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 125

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	4
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
54	

Project strong points:

1) Very good and timely project - China banned plastic. Even if ban is lifted later after negotiations, there will be sweeping changes in international as well as domestic travel and also on regulations by each individual country during post pandemic period. 2) Estimated production of 6.8 barrels of gasoline / ton, and 330 days / year, Pay Back Time 5.5 years (Suggestion: Consider if these figures should be recalculated considering changes in post pandemic America, regulations and social guidelines) 3) Estimated sale price \$1.75/gallon (Suggestion: add some text to say if this is the final price after the cost of processing unfinished gasoline). 4) Using hydrogen as fluidizing agent: (Suggestion: add text indicating safety precautions hydrogen as flammable.) 5) CO & CO2 into atmosphere: Need to calculate amount and ensure it is less than the allowable limits per Federal/State Code.

Areas for improvement:

None. Everything looks fine. (My suggestions indicated above can be considered as project enhancement). I think the Project Advisor might be able to suggest any improvement, if there are any such areas, because Advisor was involved from initiation of the project through completion. Since I am doing only an overview of the final project, I may not be able to identify areas for improvement.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 135

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
48	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 142

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
57	

Project strong points:

Realistic approach rolling out via a pilot plant for a practical problem; great understanding of handling various other plastics/material in the inlet stream. Great job.

Areas for improvement:

Would have appreciated more detail on avoiding olefins as that sounded like a potential larger concern

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 154

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
40	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 180

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
51	

Project strong points:

Very clear abstract. Good presentation overall. I like how you guys have equipment pricing in the appendix as well as NPV table. The presentation skills are good. All team members participated and had a smiling face. This project can reduce the yearly HDPE waste to a small percentage and also help producing good amount of gasoline and cater to the ever growing oil needs.

Areas for improvement:

Now that the oil prices are to their all time low due to the COVID19-- your revenue model is not relevant anymore. But jokes apart, I would like to see the time required for the initial setup of the equipments and the plant. HDPE is a type of plastic from the 419 millions in 2019. What is the percentage of HDPE in total waste? And how do you segregate HDPE from rest plastic types. I see that you are outsourcing this segregation. What are the costs involved? Would a team member be fit to check for any gas leaks or would you be hiring an expert for that'?

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 234

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	4
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
56	

Project strong points:

This is an innovate project that is needed. Plastic waste is a large amount of our waste not only here but all over the world

Areas for improvement:

Need to consider where the light gases are going and how they can be purified for eventual sale or re-use.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.16 **Category:** Environmental Engineering

Project Title: Plastics to Power: Producing Fuel

Judge ID#: 240

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
48	

Project strong points:

The material is well organized. Even for someone with a limited knowledge of chemical engineering, I was able to grasp the important concepts.

Areas for improvement:

Lots of theory, which can be good, but I would have liked to see a little more on what is currently being done and some real world numbers. The project could be brought to the real world a little better.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.2 **Category:** Environmental Engineering

Project Title: Sweeten the Planet with Polylactic Acid (PLA)

Judge ID#: 125

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	2
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
50	

Project strong points:

1) Environmental consideration is very well considered. (one million plastics around the world and 60% plastic ends in landfills). 2) Design and process flow diagram is impressive and technically correct. 3) Good design and engineering considerations and concepts of chemical engineering principles.

Areas for improvement:

1) Identify reasons for Considering only petroleum-based plastics and identify reasons for not including other types. 2) Why Shanghai, China was considered for building this facility? Building in Mexico is also cheaper and closer to USA. Any consideration was made due to the impact of COVID-19 outbreak in China? 3) Design alternatives were not discussed in this project. (Like building in places like Mexico or Canada or in USA itself???) 4) Project practicability in view of engineering standards/practice is not discussed. Can USA standards be implemented in China? Is the finished product PLA in China will be approved by American Regulating bodies> This should be discussed. If final product is not usable in USA, then even the best design is no good..

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.2 **Category:** Environmental Engineering

Project Title: Sweeten the Planet with Polylactic Acid (PLA)

Judge ID#: 135

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
47	

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.2 **Category:** Environmental Engineering

Project Title: Sweeten the Planet with Polylactic Acid (PLA)

Judge ID#: 142

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
59	

Project strong points:

Great Q&A session addressing some of the items not discussed in the presentation. Discussing alternate sources for glucose was very helpful in understanding potential improvements. Great detail in the chemical process.

Areas for improvement:

Would like to see a comparison to the market in terms of purity, profit/revenue per Ton. Additional sensitivity analysis regarding labor and utilities (especially steam) would be helpful.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.2 **Category:** Environmental Engineering

Project Title: Sweeten the Planet with Polylactic Acid (PLA)

Judge ID#: 154

Department: CHE

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
44	

Project strong points:

Strong understanding of process

Areas for improvement:

Needed more discussion of alternate and comparison to current methods.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 137

Project ID: CHE.3 Category: Bioprocess Engineering

Department: CHE

Project Title: Processing Green to be Green – Production of Renewable Diesel from Algae

Technical Quality

1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3

Presentation Quality

1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	2
6. Quality, effectiveness, and use of visual aids	3

Total Overall Score: 43

Project strong points:

I like how you thought about the 50cent bonus, and that you would be near pipelines to collect products and recycles. I think you did think through flammability hazard.

Areas for improvement:

When you describe the steps of your process, describe conditions and products. At first, I thought the only product was Biodiesel. It turns out there are actually four products. Products are not mentioned in the abstract. I think these products help your economics as much as the 50 Cent bonus does. You mention car fleets in the abstract, but told me that the demand will be in Asia for powering plant machinery.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 201

Project ID: CHE.3 Category: Bioprocess Engineering

Department: CHE

Project Title: Processing Green to be Green – Production of Renewable Diesel from Algae

Technical Quality	
1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	4
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	5
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
49	

Project strong points:

Good choice of feedstock and understanding of the technical aspects of the process. Excellent use of illustrations, not just putting them up without describing them or decorating the slide, but using them as visual aids to what was being vocally conveyed.

Areas for improvement:

Is the novelty of the project just the implementation of a known process for some company? The process is from Honeywell UOP, and algae have been a feedstock choice for some years now.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 146

Project ID: CHE.4 Category: Petrochemical Engineering

Department: CHE

Project Title: Blue in the EU: Production of Blue Hydrogen for a Greener Future

Technical Quality

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4

Presentation Quality

1. General professionalism and oral communication	5
2. Knowledge of background material	2
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	4

Total Overall Score: 46

Project strong points:

Well done.. THis is almost pilot plant size.

Areas for improvement:

A little more research in alternatives would be useful

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.4 **Category:** Petrochemical Engineering

Judge ID#: 197

Department: CHE

Project Title: Blue in the EU: Production of Blue Hydrogen for a Greener Future

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	3
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	3
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
50	

Project strong points:

Project was very inovative and the research into why this process would be useful was very well grounded. All components were stepped through thoughtly and the presentation was excellent for students of their level.

Areas for improvement:

Many challanges of why similar processes have not been globally embrassed were not address (natural gas use, catalyst selection ect...). Good for audiances open to these new ideas but could be better for those who challenge the validity of these concepts

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.5 **Category:** Environmental Engineering

Judge ID#: 125

Department: CHE

Project Title: A Bridge to Renewable Energies: Carbon Dioxide Methanation

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	2
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
54	

Project strong points:

1) Project - knowledge of background material and the idea of purchasing CO₂ from existing power plant (Mountainview) and selling it back to grid or at a better price. 2) Technically, the process flow diagram and Block diagram is very good impressive. The chemical process is very well presented and the reactions are also looking very clearly. 3) Safety Analysis has covered well from flammable point considering CO₂ and Methane.(storage in Tanks)

Areas for improvement:

1) More investigation is suggested for building Methanation Plant in California as it was recognized in the presentation that it is not legalized. (Gray Area - need to elaborate and explain gray area).Potential failure to comply with the Legal requirements must be discussed. 2) In the presentation safety issue due to ignition is discussed but structural integrity of the equipment is not discussed. In California, seismic effects are imminent, equipment must be well supported for all the loading conditions including earthquake. 3) Proposed Plant is operating at a loss. Investors will be hesitant to fund. The project should be feasible for construction. Suggestion: Do a cost/Benefit analysis and show some benefit in terms of environment protection, use of renewable energy, etc and assign some dollar value to those benefits. Overall calculation will show some profit instead of a loss.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.5 **Category:** Environmental Engineering

Judge ID#: 135

Department: CHE

Project Title: A Bridge to Renewable Energies: Carbon Dioxide Methanation

Technical Quality

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints	2
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	3

Presentation Quality

1. General professionalism and oral communication	3
2. Knowledge of background material	3
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3

Total Overall Score: **38**

Project strong points:

Areas for improvement:

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 142

Project ID: CHE.5 Category: Environmental Engineering

Department: CHE

Project Title: A Bridge to Renewable Energies: Carbon Dioxide Methanation

Technical Quality

1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	5

Presentation Quality

1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5

Total Overall Score: 56

Project strong points:

Practical understanding of project viability and hurdles. Great explanation of the Carbon Dioxide Methanation process.

Areas for improvement:

With Time of Day utility pricing, an evaluation of necessary conditions and limitations to project viability appear to be the next logical step.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 154

Project ID: CHE.5 **Category:** Environmental Engineering

Department: CHE

Project Title: A Bridge to Renewable Energies: Carbon Dioxide Methanation

Technical Quality

1. Identification of project objective(s)	3
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	3

Presentation Quality

1. General professionalism and oral communication	4
2. Knowledge of background material	3
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4

Total Overall Score: 39

Project strong points:

Good presentation materials

Areas for improvement:

Needed to discuss alternatives more.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.6 **Category:** Bioprocess Engineering

Judge ID#: 137

Department: CHE

Project Title: Subdued Crude Paving the Way for Less-Lewd Renewed Car Food

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	5
5. Project innovation	5
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	5
Total Overall Score:	
56	

Project strong points:

I loved how you prepared for this presentation. You used the laser pointer, and were familiar enough with your slides to move right to the one needed. Everyone on the team seemed to be equally knowledgeable about the material. You had the best consideration and presentation of hazards to employees and environment (not just process) that I have seen.

Areas for improvement:

Your innovation is in saving on the heat exchanger network. That is not mentioned in the abstract. I am glad you pointed it out to me in the presentation. You gave a number of the savings due to this analysis. I would have liked to see the impact on the sensisitivity

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 201

Project ID: CHE.6 Category: Bioprocess Engineering

Department: CHE

Project Title: Subdued Crude Paving the Way for Less-Lewd Renewed Car Food

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	5
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	5
3. Organization and quality of presentation materials	5
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	5
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
56	

Project strong points:

Great economic analysis! Got the full picture. Good thing that you address treatment of your wastewater.

Areas for improvement:

Some slides need better referencing. A little bit more consideration of safety issues would be nice.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 137

Project ID: CHE.7 Category: Bioprocess Engineering

Department: CHE

Project Title: Tough Wrinkles? Bring in the Heavy Weight, Sodium Hyaluronate

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	5
4. Overall fulfillment of project objectives	5
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	0
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
44	

Project strong points:

I am trying not to be biased here. I think you truly understand the process, and could communicate on the fly without referring to the slide or to a paper. You understood the Strep hazard and articulated it. I think your economics do correctly cover the plant, and show the weakness of having so much capital without further refinement of the building dimensions and space use.

Areas for improvement:

I know there was an abstract written, but it is not here. I think your knowledge or perhaps the presentation of, the hazards of the Ethanol was weak.

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 201
Department: CHE

Project ID: CHE.7 **Category:** Bioprocess Engineering

Project Title: Tough Wrinkles? Bring in the Heavy Weight, Sodium Hyaluronate

Technical Quality	
1. Identification of project objective(s)	4
2. Evaluation of design alternatives with realistic constraints	5
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	4
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
48	

Project strong points:

Good explanation as to why using fed batch vs. continuous fermentation

Areas for improvement:

Slides are not standalone materials, they are visual aids to further explain or expand what is being said. Neither are they meant to be fully read.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.8 **Category:** Bioprocess Engineering

Project Title: Biodiesel Production from Soybean Oil

Judge ID#: 137

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	3
5. Project innovation	3
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	5
2. Knowledge of background material	3
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	4
5. Group dynamics/member participation	3
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
44	

Project strong points:

I like how you thought about placement of the facility, and thought about pricing and availability of feedstock. You understood your pricing and cost sensitivity, and accurately displayed it, in a clear way, in the presentation. You could speak to it on the fly.

Areas for improvement:

I don't think you thought through your hazards. You thought through hazards to the process working well, but not hazards to employees and the community. It is not a given that your sewage will be OK to go the municipal treatment plant. I don't think you thought through the hazards due to the large amount of Caustic you would be using for your plant. I don't feel like you had a sense about how big your equipment or your plant was.

EXPO 2020 PROJECT JUDGING RESULTS

Project ID: CHE.8 **Category:** Bioprocess Engineering

Project Title: Biodiesel Production from Soybean Oil

Judge ID#: 201

Department: CHE

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	4
3. Consideration of factors influencing the design	4
4. Overall fulfillment of project objectives	5
5. Project innovation	2
6. Project practicality in view of contemporary engineering standards/practice	5
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	4
4. Project description (see project abstract in program book)	5
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	3
Total Overall Score:	
49	

Project strong points:

Areas for improvement:

Perhaps another recommendation you could have is a pilot scale study

EXPO 2020 PROJECT JUDGING RESULTS

Judge ID#: 197

Project ID: CHE.9 **Category:** Petrochemical Engineering

Department: CHE

Project Title: From Raw Materials to High Demand Household Products-The Production of Polypropylene

Technical Quality	
1. Identification of project objective(s)	5
2. Evaluation of design alternatives with realistic constraints	3
3. Consideration of factors influencing the design	3
4. Overall fulfillment of project objectives	5
5. Project innovation	4
6. Project practicality in view of contemporary engineering standards/practice	4
Presentation Quality	
1. General professionalism and oral communication	4
2. Knowledge of background material	4
3. Organization and quality of presentation materials	3
4. Project description (see project abstract in program book)	3
5. Group dynamics/member participation	4
6. Quality, effectiveness, and use of visual aids	4
Total Overall Score:	
46	

Project strong points:

Good background knowledge and detailed engineering sensibility.

Areas for improvement:

Many typical challenges with process (temperature, pressure, coking ect) not addressed.